

추가과제1. MIFlow 설치 및 실행

1. docker 프로그램 설치

> <https://www.docker.com/>

2. pyspark-notebook docker image를 다운로드 및 실행

```
> docker run -p 5000:5000 -v mydir/myfolder:/home/jovyan/work/ jupyter/pyspark-notebook:latest
```

* 여기서 mydir/myfolder는 code와 data 폴더(mlflow_example.zip)가 있는 경로여야함.

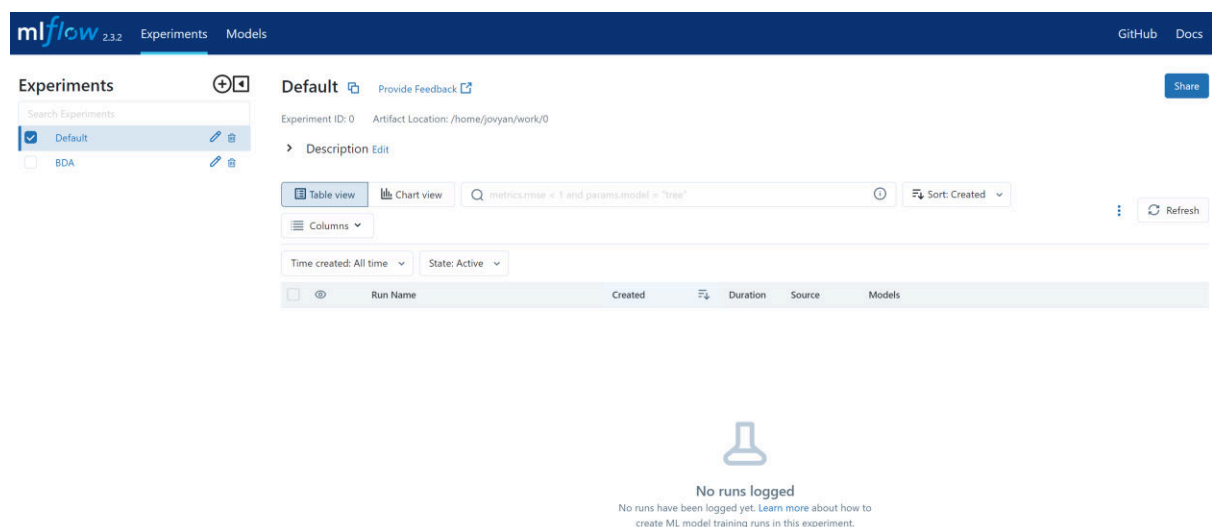
3. 해당 도커 터미널에서 다음 명령어를 통해 mlflow 설치

```
> conda install -c conda-forge mlflow
```

4. mlflow를 설치한뒤에 다음 명령어를 통해서 mlflow 서버를 실행

```
> mlflow server --backend-store-uri sqlite:///mlflow.db --default-artifact-root /home/jovyan/work -host 0.0.0.0 --port 5000 &
```

5. 웹브라우저에서 localhost:5000 입력하여 Mlflow가 정상적으로 뜨는지를 확인



6. terminal 창에서 다음 코드를 입력하여 MLFLOW 내 실행이 추적될 수 있도록 설정

> export MLFLOW_TRACKING_URI=http://localhost:5000

7. 다음 명령어를 수행하여 랜덤 포레스트 분류기를 여러 차례 실행(runs)

```
> spark-submit --master local[*] /home/jovyan/work/code/mlflow_example.py 32 5
> spark-submit --master local[*] /home/jovyan/work/code/mlflow_example.py 16 3
> spark-submit --master local[*] /home/jovyan/work/code/mlflow_example.py 16 5
> spark-submit --master local[*] /home/jovyan/work/code/mlflow_example.py 32 10
```

8. mlflow ui에 다시 접속하여 Experiments에 수행된 내용을 확인하고, 모델을 등록(Register Model)

< 모델 선택후, 우측 하단에 register model 선택 >

수행한 모델에는 ID값이 할당되어 있음 (runs:/id)

The screenshot displays the MLflow web interface. At the top, there's a navigation bar with 'mlflow 2.3.2', 'Experiments', and 'Models' tabs. The 'Experiments' tab is active, showing a list of experiments. The experiment 'resilient-colt-646' is selected, showing details like Run ID, Date, Source, User, Duration, and Status. Below this, there are expandable sections for Description, Parameters, Metrics, Tags, and Artifacts. The 'Artifacts' section is expanded, showing a file tree for 'spark-model' containing 'sparkml', 'MLmodel', 'conda.yaml', 'python_env.yaml', and 'requirements.txt'. The 'MLflow Model' section is also visible, providing instructions on how to use the model and a 'Register Model' button. The 'Model schema' section shows a table with 'Name' and 'Type' columns, and the 'Make Predictions' section shows a code snippet for loading and using the model.

Name	Latest Version	Staging	Production	Last Modified	Tags
BDA	Version 1	-	-	2023-05-23 16:49:10	-

7. terminal에서, 모델 폴더(아래 빨간색 부분)를 확인하여 다음 명령어를 실행

```
> spark-submit --master local[*] predict_spark.py
/home/jovyan/work/1/1c408f504de148209598df7be97f31ba/artifcats/spark-model
```

```
23/05/23 07:53:56 INFO CodeGenerator: Code generated in 9.91281 ms
+-----+-----+-----+-----+
|rawPrediction|probability|label|features|
+-----+-----+-----+-----+
|[18.36831020301444,1.6316897969855622]| [0.9184155101507219,0.0815844898492781]| 0 |[19.0,626.0,15.0,117.0,1.0,-1.0,0.0]|
|[18.36831020301444,1.6316897969855622]| [0.9184155101507219,0.0815844898492781]| 0 |[19.0,5368.0,4.0,77.0,6.0,-1.0,0.0]|
|[15.143686586236463,4.856313413763536]| [0.7571843293118231,0.24281567068817683]| 1 |[20.0,76.0,18.0,639.0,2.0,-1.0,0.0]|
|[18.36831020301444,1.6316897969855622]| [0.9184155101507219,0.0815844898492781]| 0 |[20.0,67.0,19.0,387.0,1.0,-1.0,0.0]|
|[18.36831020301444,1.6316897969855622]| [0.9184155101507219,0.0815844898492781]| 0 |[20.0,130.0,4.0,75.0,3.0,-1.0,0.0]|
+-----+-----+-----+-----+
only showing top 5 rows
```

과제 제출 내용:

0. 학번 / 이름

1. (mlflow_example.py) Mlflow 적어도 5개의 runs를 수행한 결과화면 (5개를 초과/미만해서 안되며, 삭제해야함)

2. (predict_spark.py) 7의 그림과 같이 예측된 값에 대해서 출력화면

주의사항:

- 프로그램 오류는 개별적으로 해결해야 함.
- 타 학생의 스크린샷을 사용하는 경우, 이유를 불문하고 0 점 처리함